Simulation-Generated Data for Agentic Course of Action Development and World Models

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ABSTRACT

Data-driven military decision support requires generative planning to be grounded in the physical and operational consequences of proposed actions. While Large Language Models (LLMs) can rapidly synthesize doctrinally fluent courses of action (COAs), they lack intrinsic world models to evaluate operational constraints such as terrain, platform dynamics, timing, logistics, and attrition. We present a decision support architecture that grounds a multi-agent, multi-modal LLM planning pipeline, aligned with the Norwegian Army planning and decision-making process, within a closed-loop constructive simulation. Rather than utilizing simulation solely for post-hoc validation, our approach integrates it directly into iterative plan refinement.

The pipeline begins with terrain analysis agents leveraging vision-language models and spatial services to build structured military geography. This feeds a HybridRAG (Hybrid Retrieval-Augmented Generation) layer combining a vector database of doctrine and field manuals with a knowledge graph of the operational environment, order of battle, and unit command structures. By traversing relationships across this graph, such as verifying whether adjacent units can mutually support one another, the system identifies and filters doctrinally unsound maneuvers before simulation execution. For physical grounding, candidate COAs are translated into the C2SIM (Command and Control Systems to Simulation Systems Interoperation) standard to task the SWAP (Simulation-Supported Wargaming for Analysis of Plans) constructive simulation. Machine-readable simulation data—including routes, force attrition, timing, and logistical expenditures—is fed back to the agents for iterative refinement and archived within the knowledge layer to accumulate operational experience, while also allowing human planners to inject critiques and command constraints into the loop.

Beyond plan evaluation, this closed-loop architecture serves as a data generation engine. The resulting state-action-outcome trajectories provide the empirical foundation to train a battlespace world model. Operating as a fast surrogate, this model quickly screens candidate plans by predicting their operational outcomes, reserving high-fidelity simulation for the most promising alternatives. The core agentic pipeline and HybridRAG components are implemented and validated, the standards-based simulation loop is undergoing integration, and the world model represents an active research direction for scalable, grounded tactical AI.

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